FIITJEE
ALL INDIA TEST SERIES
JEE (Advanced)-2024
CONCEPT RECAPITULATION TEST – IV
PAPER –2
TEST DATE: 01-05-2024

Time Allotted: 3 Hours **Maximum Marks: 180**

General Instructions:

- The test consists of total 51 questions.
- Each subject (PCM) has 17 questions.
- This question paper contains **Three Parts.**
- **Part-I** is Physics, **Part-II** is Chemistry and **Part-III** is Mathematics.
- Each **Part** is further divided into **Three Sections**: **Section-A, Section-B & Section-C**.
  Section – A *(01 – 04, 18 – 21, 35 – 38)*: This section contains **TWELVE (12)** questions. Each question has **FOUR** options. **ONLY ONE** of these four options is the correct answer.
  Section – A *(05 –07, 22 – 24, 39 – 41)*: This section contains **NINE (09)** questions. Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is(are) correct answer(s).
  Section – B *(08 – 13, 25 – 30, 42 – 47):* This section contains **EIGHTEEN (18)** numerical based questions. The answer to each question is a **NON-NEGATIVE INTEGER VALUE**.
  Section – C *(14 –17, 31 – 34, 48 – 51)*: This section contains **SIX (06) paragraphs.** Based on each paragraph, there are **TWO (02)** questions of numerical answer type. The answer to each question is a **NUMERICAL VALUE (XXXXX.XX).** If the numerical value has more than two decimal places, truncate/round-off the value to **TWO** decimal places.

MARKING SCHEME

Section – A (Single Correct): Answer to each question will be evaluated according to the following marking scheme:

Full Marks	:	+3	If ONLY the correct option is chosen.
Zero Marks	:	0	If none of the options is chosen (i.e. the question is unanswered);
Negative Marks	:	–1	In all other cases.

Section – A (One or More than One Correct): Answer to each question will be evaluated according to the following marking scheme:

Full Marks	:	+4	If only (all) the correct option(s) is (are) chosen;
Partial Marks	:	+3	If all the four options are correct but ONLY three options are chosen;
Partial marks	:	+2	If three or more options are correct but ONLY two options are chosen and both of which are correct;
Partial Marks	:	+1	If two or more options are correct but ONLY one option is chosen and it is a correct option;
Zero Marks	:	0	If none of the options is chosen (i.e. the question is unanswered);
Negative Marks	:	–2	In all other cases.

Section – B: Answer to each question will be evaluated according to the following marking scheme:

Full Marks	:	+4	If ONLY the correct integer is entered;
Zero Marks	:	0	Question is unanswered;
Negative Marks	:	0	In all other cases.

Section – C: Answer to each question will be evaluated according to the following marking scheme:

Full Marks	:	+3	If ONLY the correct integer is entered;
Zero Marks	:	0	Question is unanswered;
Negative Marks	:	0	In all other cases.

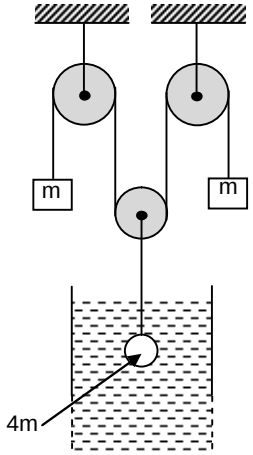
Physics

PART – I

SECTION – A
(One Options Correct Type)

*This section contains **FOUR (04)** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is the correct answer.*

1. A spherical ball of mass 4m, density σ and radius r is attached to a pulley-mass system as shown in figure. The ball is released in a liquid of coefficient of viscosity η and density $\rho\left(<\frac{\sigma}{2}\right)$. If the length of the liquid column is sufficiently long, the terminal velocity attained by the ball is given by (assume all pulleys to be massless and string as massless and inextensible):

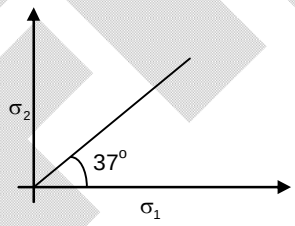


(A) $\frac{2}{9}\frac{r^2(2\sigma-\rho)g}{\eta}$ (B) $\frac{2}{9}\frac{r^2(\sigma-2\rho)g}{\eta}$

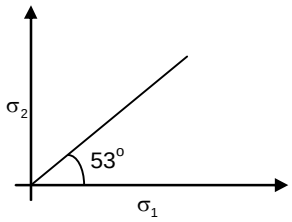
(C) $\frac{2}{9}\frac{r^2(\sigma-4\rho)g}{\eta}$ (D) $\frac{2}{9}\frac{r^2(\sigma-3\rho)g}{\eta}$

2. Two sphere of radii R_1 and R_2, separated by a large distance, are joined by a fine wire, are raised to a potential V. If σ_1 and σ_2 represent the respective surface charge densities of the spheres, then for $\frac{R_1}{R_2}=\frac{3}{4}$ we have:

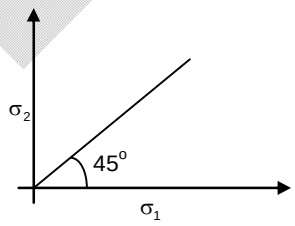
(A)



(B)

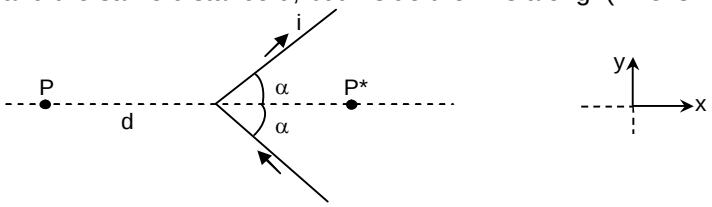


(C)



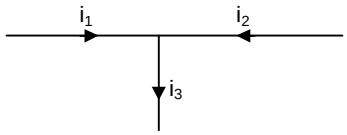
(D) None of these

3. The direction of field B at point P* symmetric to P with respect to the vertex, i.e., along the axis and the same distance d, but inside the V is along: (wire is in x-y plane)



(A) positive z-axis
(B) negative x-axis
(C) negative z-axis
(D) negative y-axis

4. In the given figure if $i_1 = 3\sin\omega t$ and $i_2 = 4\cos\omega t$, then i_3 is:

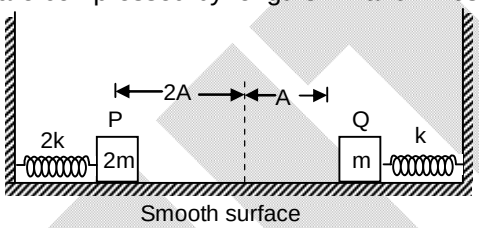


(A) $5\sin\left(\omega t + 53^o\right)$
(B) $5\sin\left(\omega t + 37^o\right)$
(C) $5\sin\left(\omega t + 45^o\right)$
(D) $5\cos\left(\omega t + 53^o\right)$

SECTION – A
(One or More than one correct type)

*This section contains **THREE (03)** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).*

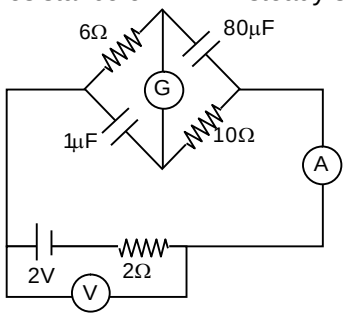
5. Two objects P and Q of masses 2m and m are connected to two springs of spring constant 2k and k respectively. Other ends of springs are connected to rigid walls as shown. Initially springs are compressed by lengths 2A and A respectively, and then released:



(A) Time period of P and Q remains same before and after the collision irrespective of the fact whether collision between masses is perfectly elastic or perfectly inelastic.
(B) Timer period of P and Q remains same before and after the collision only if collision between the objects is perfectly elastic.
(C) Time period of P and Q remains same before and after the collision only if collision between the objects is perfectly inelastic.
(D) If collision is perfectly inelastic, amplitude of SHM of combined mass is A.

6. The position of a particle moving along x-axis is given by $x = 3t^2 - t^2$; where x is in m and t is in sec. Consider the following statements:
(A) Displacement of the particle after 4s is 16 m
(B) Distance travelled by the particle up to 4s is 24 m
(C) Displacement of the particle after 4s is (-16) m
(D) Distance covered by the particle up to 4s is 22 m

7. The ammeter (A) and voltmeter (V) in the given circuit are ideal but the galvanometer has a resistance of 2Ω. In steady state,



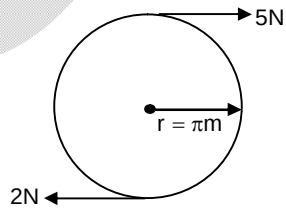
(A) No current flows in (G)
(B) 0.1 A current flows in (G)
(C) Charge on 1 μF capacitor is 0.8 μC
(D) Terminal potential difference across battery is 1.8 V

SECTION – B

(Numerical Answer Type)

*This section contains **SIX (06)** Numerical based questions. The answer to each question is a **NON-NEGATIVE INTEGER VALUE**.*

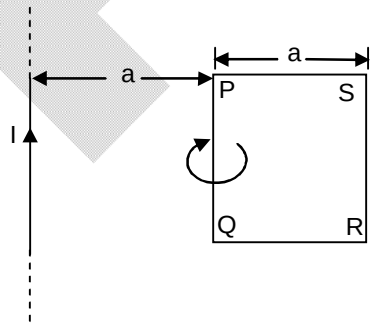
8. Figure shows a cylinder of mass $\frac{2}{7}$kg and radius π m. Two antiparallel constant forces 5 N and 2N act on it tangentially as shown in figure. Find the square of velocity $(m/s)^2$ of center of cylinder as it rotates by 180°. Neglect gravity and friction. [Take $\pi^2 = 10$]



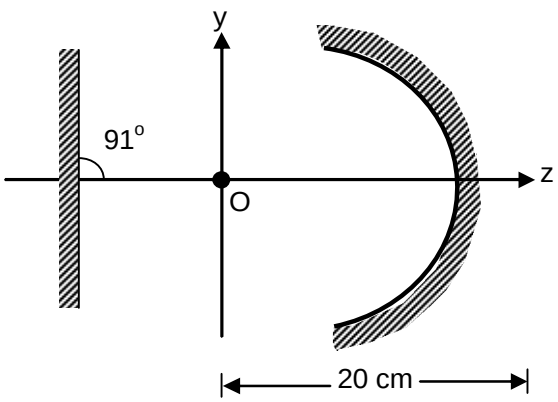
9. Six points masses of mass m each are at the vertices of a regular hexagon of side l. The force on any one of the masses is given by $F = \frac{Gm^2}{l^2}\left(\frac{1}{x}+\frac{1}{\sqrt{y}}+\frac{1}{z}\right)$. Then find $x + y - z$. (Take: $x < y < z$).

10. A neutron is scattered through ($\equiv$ deviation from its old direction) θ degree in an elastic collision with an initially stationary deutron. If the neutron loses 2/3 of its initial KE to the deutron, then find the value of θ (in degree). (In atomic mass unit, the mass of a neutron is 1 u and mass of a deutron is 2 u)

11. A square frame of resistance In (2) Ω and side a = 20 cm and a long straight wire carrying a current I = 10 amp are located in the same plane. The frame is rotated through an angle of 120° about the side PQ. The amount of charge flown through the loop during this time is $q \times 10^{-7}$ coulomb, find the value q.



12. A point object is placed at the center of curvature of a concave mirror (taken as origin). A plane mirror is also placed at a distance of 10 cm from the object as shown. Consider two reflection first at plane mirror and then at concave mirror. If (x_0, y_0) is the co-ordinate of the image thus formed, then find $\frac{\pi x_0}{y_0}$.



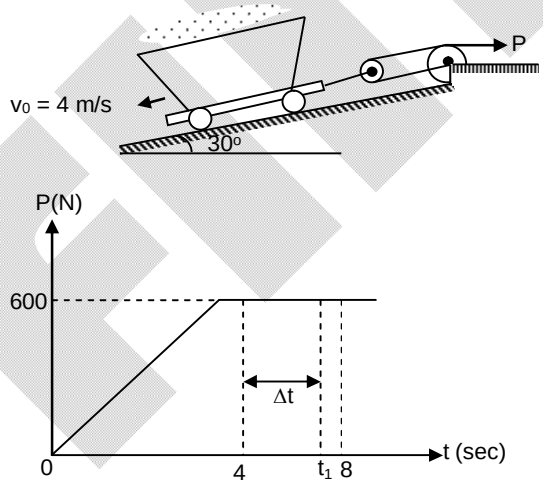
13. The shortest wavelength of the Bracket series of a hydrogen-like atom (atomic number = Z) is the same as the shortest wavelength of the Balmer series of hydrogen atom. Find the value of Z.

SECTION – C
(Numerical Answer Type)

*This section contains **TWO (02) paragraphs.** Based on each paragraph, there are **TWO (02)** questions of numerical answer type. The answer to each question is a **NUMERICAL VALUE (XXXXX.XX).** If the numerical value has more than two decimal places, truncate/round-off the value to **TWO** decimal places.*

Paragraph for Question Nos. 14 and 15

The loaded 150 kg skip is rolling down the incline at 4 m/s when a force P is applied to the cable as shown at time t = 0. The force P is increased uniformly with the time until it reaches 600 N at t = 4 s, after which it remains constant at this value.



14. The time t_1 at which the skip reverses its direction is nearly _____ sec.

15. The velocity v of the skip at t = 8 s (treating the skip as a particle) is _____ m/s.

Paragraph for Question Nos. 16 and 17

A quantity of an ideal monoatomic gas consists of n moles initially at temperature T_1. The pressure and volume are then slowly doubled in such a manner so as to trace out a straight line on a P-V diagram.

16. For this process, the ratio $\frac{W}{nRT_1}$ is equal to (where W is work done by the gas) _____.

17. For the same process, the ratio $\frac{Q}{nRT_1}$ is equal to (where Q is heat supplied to the gas) _____.

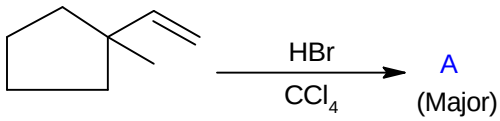
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Chemistry

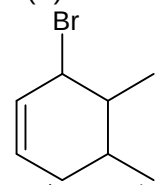
PART – II

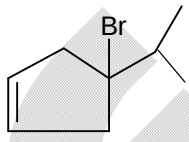
SECTION – A
(One Options Correct Type)

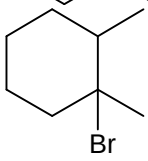
*This section contains **FOUR (04)** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is the correct answer.*

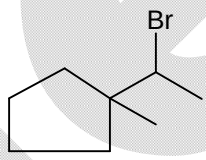
18. 

Product (A) is:

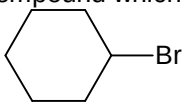
(A) 

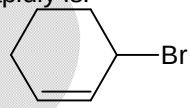
(B) 

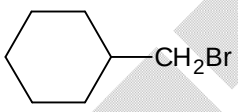
(C) 

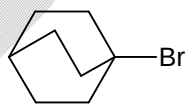
(D) 

19. The compound which undergoes SN^1 reaction most rapidly is:

(A) 

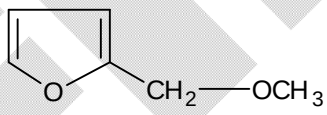
(B) 

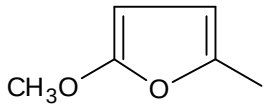
(C) 

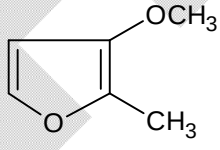
(D) 

20. 

Which of the following products can be obtained from above reaction?

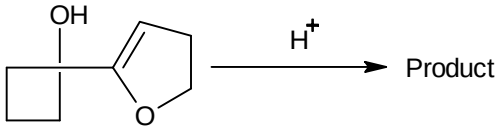
(A) 

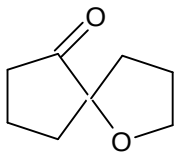
(B) 

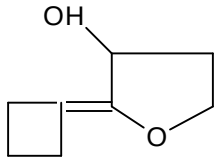
(C) 

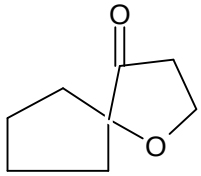
(D) All of these

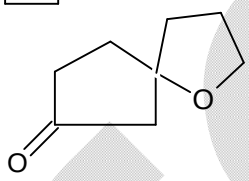
21. Identify the major product,



(A) 

(B) 

(C) 

(D) 

SECTION – A
(One or More than one correct type)

*This section contains **THREE (03)** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).*

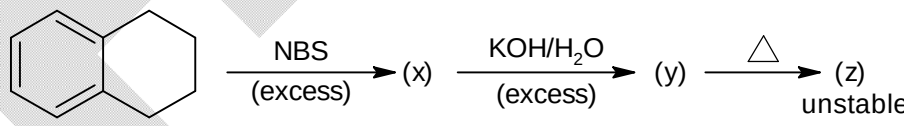
22. In which of the following option(s) the dipole moment of the left side molecule is greater than that of the right side molecule?
(A) NH_3, NF_3 (B) $SCl_2, BeCl_2$
(C) SF_4, CF_4 (D) PF_5, IF_5

23. Saturated solution of SO_2 is heated at 150° in a closed container. The product obtained is treated with $BaCl_2$ solution. What is/are the observation(s)?
(A) no. ppt. (B) white turbidity
(C) evolution of SO_2 (D) white ppt.

24. Which of the following compound (s) will not dissolve in an alkali solution after it has undergoes reaction with Hinsberg's reagent?
(A) Diethyl amine (B) Isopropyl amine
(C) N-methyl ethanamine (D) N, N-dimethyl aniline

SECTION – B
(Numerical Answer Type)

*This section contains **SIX (06)** Numerical based questions. The answer to each question is a **NON-NEGATIVE INTEGER VALUE**.*

25. 

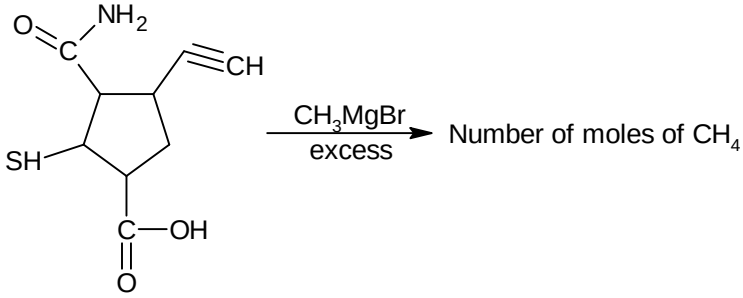
The sp^2 – hybridised carbon atoms present in one molecule of the end product z is:

26. Find $10 \times \log_{10} K$ for the reaction at 298 K. $2H^+(aq) + D_2(g) \rightleftharpoons H_2(g) + 2D^+(aq)$

Given $E^0_{D^+/D_2} = -2.955$ mV

Take $\left(\frac{RT2.303}{F}\right) = 0.0591$

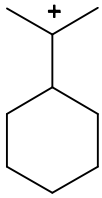
27. What is the number of moles of CH_4 formed in the given reaction?



28. For a reaction $X + Y \rightleftharpoons 2Z$, 1.0 mol of X, 1.5 mol of Y and 0.5 mol of Z were taken in a 1 L vessel and allowed to react. At equilibrium, the concentration of Z was 1.0 mol L^{-1}. The equilibrium constant of the reaction is $\frac{x}{15}$. The value of $\frac{x}{15}$ is approximately (nearest integer value).

29. If 75% of a first order reaction was completed in 90 minutes, 60% of the same reaction would be completed in approximately (in minutes) is x. Then the value of $\frac{x}{10}$ is _____. $(\text{Take} : \log 2 = 0.30;\ \log 2.5 = 0.40)$

30. The total number of contributing structures showing hyperconjugation (involving C – H bonds) for the following carbocation is



SECTION – C
(Numerical Answer Type)

*This section contains **TWO (02) paragraphs.** Based on each paragraph, there are **TWO (02)** questions of numerical answer type. The answer to each question is a **NUMERICAL VALUE (XXXXX.XX).** If the numerical value has more than two decimal places, truncate/round-off the value to **TWO** decimal places.*

Paragraph for Question Nos. 31 and 32

The freezing point depression of a 0.109 M aqueous solution of formic acid is –0.21°C . The percentage dissociation of formic acid and equilibrium constant for the reaction of its dissociation in water is a and $b \times 10^{-4}$ respectively. (K_f for water = 1.86 K.kg/mol, let Molarity = molality for above solution)

31. The value of a is

32. The value of b is

Paragraph for Question Nos. 33 and 34

In 1886, the French chemist, Francois Raoult after a series of experiments on a number of solvents including water, benzene and ether, succeeded in establishing a relationship between the lowering of vapour pressure of a solution and the mole fraction of the non volatile solute.

$P_1 = X_1 P_1^0$

where P_1 is vapour pressure of solvent in an ideal solution and P_1^0 is vapour pressure of a pure solvent. X_1 mole fraction of solvent and X_2 mole fraction of solute.

$X_1 + X_2 = 1$

$P_1 = (1 - X_2)P_1^0$

$\frac{P_1^0 - P_1}{P_1^0} = X_2$

The relative lowering of vapour pressure of a solution containing a non-volatile solute is equal to the mole fraction of solute present in the solution.

33. The vapour pressure of pure liquid X and pure liquid Y are 32 and 102 mm of Hg respectively at 400 K. A liquid mixture of X and Y is composed of 1 mole each. What will be the pressure, when 1 mole of the mixture has been vapourized.

34. A solution of non-volatile solute AX in a mixture of solvent A and Solvent B has a vapour pressure of 40 mm of Hg at 20° C and 68 mm of Hg at 30° C. The vapour pressure of pure solvent (A) is 15 mm of Hg at 20° C and 34 mm of Hg at 30° C and the vapour pressure of pure solvent (B) is 85 mm of Hg at 20° C and 136 mm Hg at 30° C. Find the sum of mole fraction of solvent A and solute.

Mathematics

PART – III

SECTION – A
(One Options Correct Type)

*This section contains **FOUR (04)** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is the correct answer.*

35. Solution of the differential equation $x^2ydy+\left(x^2\left(x+y\right)-y^2\left(2x+y\right)\right)dx=0$ is given by (here C is constant of integration)

(A) $\ln\left(\frac{C(y-x)}{x^4(y+x)}\right)=\frac{2x}{x+y}$ (B) $\ln\left(\frac{C(y+x)}{x^4(y+2x)}\right)=\frac{2x}{x+y}$

(C) $\ln\left(\frac{C(y-2x)}{x^4(y+x)}\right)=\frac{2x}{x+y}$ (D) $\ln\left(\frac{C(y-x)}{x^3(y+x)}\right)=\frac{2x}{x+y}$

36. Let $f: R \to R$, $f(x) = x^3 + 4x - 1$ be a function whose inverse is $g(x)$, then $g''(4)$ is equal to-

(A) $-\frac{6}{343}$ (B) $-\frac{6}{49}$

(C) $\frac{1}{49}$ (D) $-\frac{3}{49}$

37. Let A be the area of the ellipse $x^2+4y^2=16$ and let S be the area of the largest circle centred at (1, 0) that can be inscribed inside the ellipse then $\frac{A}{S}$ is equal to

(A) $\frac{12}{15}$ (B) $\frac{11}{5}$

(C) $\frac{24}{11}$ (D) $\frac{16}{9}$

38. Number of real value(s) of k for which the following system of equations $kx + y = k + 1$, $2x + y = 2k - 1$, $x - ky = 3$ is consistent is equal to

(A) 0 (B) 1

(C) 2 (D) 3

SECTION – A
(One or More than one correct type)

*This section contains **THREE (03)** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).*

39. If the greater root of the equation $x^2-\left(a^2+a+1\right)x+a\left(a^2+1\right)=0$ lies between the roots of the equation $x^2-a^2x-2\left(a^2-2\right)=0$ then, the smallest positive integral value of a, is less than

(A) 2 (B) 3

(C) 4 (D) 5

40. If $f(x)$ be double differentiable function such that $|f''(x)| \le 5\ \forall\ x \in [0, 4]$ and f takes its largest value at an interior point of this interval, then the value of $\left|f'(0)\right|+\left|f'(4)\right|$ can be

(A) 18 (B) 19

(C) 20 (D) 21

41. Consider a complex number z on the argand plane satisfying $\arg\left(z^2-\omega\right)=\frac{\pi}{2}+\arg\left(z^2-\omega^2\right)$ where $\omega=e^{i\frac{2\pi}{3}}$, then identify the correct statement(s):

(A) Maximum value of $|z|$ is $\frac{\sqrt{3+\sqrt{3}}}{2}$

(B) Maximum value of $|z|$ is $\frac{\sqrt{2+2\sqrt{3}}}{2}$

(C) Maximum value of $|z-1-i||z+1+i|$ is equal to $\frac{\sqrt{17}+\sqrt{3}}{2}$

(D) Minimum value of $|z-1-i||z+1+i|$ is equal to $\frac{\sqrt{17}-\sqrt{3}}{2}$

SECTION – B

(Numerical Answer Type)

*This section contains **SIX (06)** Numerical based questions. The answer to each question is a **NON-NEGATIVE INTEGER VALUE**.*

42. Let $f(x)=\min\left(\{x^2\},\cos x\right)$ (where {.} denotes fractional part function), $x\in\left(0,\frac{\pi}{2}\right)$ and m, n denotes the number of points of discontinuity and non- derivability of $f(x)$ respectively, then (m + n) is equal to…….

43. $s=\sum_{k=0}^{\infty}\frac{2^k}{7^{2^k}+1}$ then $\frac{1}{s}$ is equal to

44. Let vertices of a parallelogram ABCD are $A(3, 1)$, $B(13, 6)$, $C(13, 21)$ and $D(3, 16)$. If a line having slope m and passing through origin divides the parallelogram into two congruent parts, then the value of $8m$ is

45. If $\tan 9\theta=\frac{3}{4},\left(\text{where} 0<\theta<\frac{\pi}{18}\right)$ then the value of (3 cosec 3θ − 4 sec 3θ) is

46. If $g(x)$ is the inverse of $f(x)$ and $f(x)$ has domain $x\in[1, 5]$ where $f(1) = 2$ and $f(5) = 10$, then the value of integration $\int_1^5 f(x)dx+\int_2^{10} g(y)dy$ is

47. A set of consecutive positive integers starting with 1 is written on a blackboard. One number is erased, and thus the A.M of the remaining numbers is found to be $35\frac{7}{17}$. The erased number is

SECTION – C
(Numerical Answer Type)

*This section contains **TWO (02) paragraphs.** Based on each paragraph, there are **TWO (02)** questions of numerical answer type. The answer to each question is a **NUMERICAL VALUE (XXXXX.XX).** If the numerical value has more than two decimal places, truncate/round-off the value to **TWO** decimal places.*

Paragraph for Question Nos. 48 and 49

Let P denotes the plane bisecting the line segment joining points $A(-4, 2, 1)$ and $B(2,-4, 3)$ perpendicularly and Q be the plane $x - y + \lambda z = 1$, where $\lambda \in R$.

48. If the angle between the planes P and Q is 45^0 then the sum of all possible values of λ is

49. If the line L with equation $\frac{x-1}{1} = \frac{y+2}{3} = \frac{7-z}{1}$ intersects the plane P at the point $R(x_0, y_0, z_0)$, then the sum $(x_0 + y_0 + z_0)$ has the value equal to

Paragraph for Question Nos. 50 and 51

Let $f(x)$ be a function such that $f(x) = e^x(2x-1) - ax + a$ where a is a parameter and $a < 1$. If there exist one and only one $x_0 \in I$ such that $f(x_0) < 0$, then the range of a is $\left[\frac{p}{qe}, r\right)$, where p, q are co-prime.

50. The value of $(p + q + r)$ is

51. The value of $\tan^{-1}(\tan p) + \cos^{-1}(\cos q)$ is